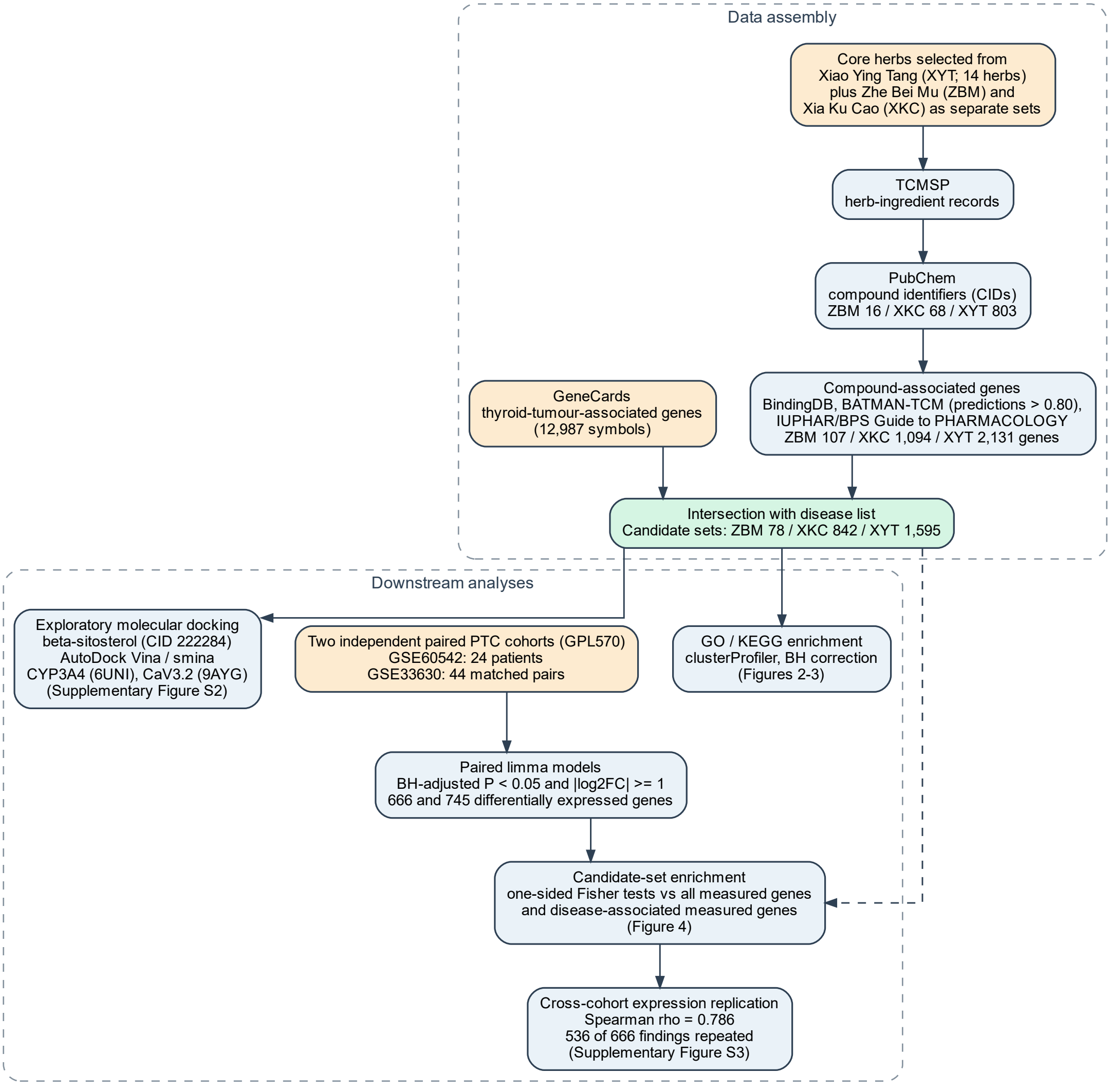
Data assembly
Core herbs selected from
Xiao Ying Tang (XYT; 14 herbs)
plus Zhe Bei Mu (ZBM) and
Xia Ku Cao (XKC) as separate sets
TCMSP
herb-ingredient records
PubChem
compound identifiers (CIDs)
ZBM 16 / XKC 68 / XYT 803
GeneCards
thyroid-tumour-associated genes
(12,987 symbols)
Compound-associated genes
BindingDB, BATMAN-TCM (predictions > 0.80),
IUPHAR/BPS Guide to PHARMACOLOGY
ZBM 107 / XKC 1,094 / XYT 2,131 genes
Intersection with disease list
Candidate sets: ZBM 78 / XKC 842 / XYT 1,595
Downstream analyses
Exploratory molecular docking
beta-sitosterol (CID 222284)
AutoDock Vina / smina
CYP3A4 (6UNI), CaV3.2 (9AYG)
(Supplementary Figure S2)
Two independent paired PTC cohorts (GPL570)
GSE60542: 24 patients
GSE33630: 44 matched pairs
GO / KEGG enrichment
clusterProfiler, BH correction
(Figures 2-3)
Paired limma models
BH-adjusted P < 0.05 and |log2FC| >= 1
666 and 745 differentially expressed genes
Candidate-set enrichment
one-sided Fisher tests vs all measured genes
and disease-associated measured genes
(Figure 4)
Cross-cohort expression replication
Spearman rho = 0.786
536 of 666 findings repeated
(Supplementary Figure S3)